Chapter 3

氧化工艺

Oxidation of Silicon

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2025年11月17日

Outline

- 3. 1 Why is oxidation needed?
- 3. 2 Basics of thermal oxidation
- 3. 3 Kinetics of thermal oxidation
- 3. 4 Other aspects of thermal oxidation
- 3. 5 Summary

3. 1 Why is oxidation of Si needed?

Ge has high μ_e, μ_h,
Ge is stable…,
…but no good oxide

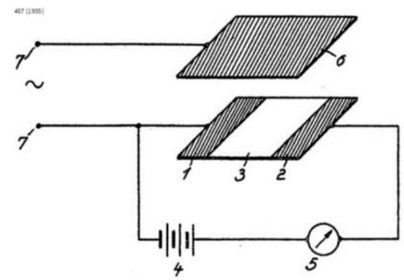
GaAs has high μ_e
and direct band…,
…but no good oxide

Probably safe to sat that history of IC would be different without good SiO_2 grown on Si wafer

Concept of FET, patent

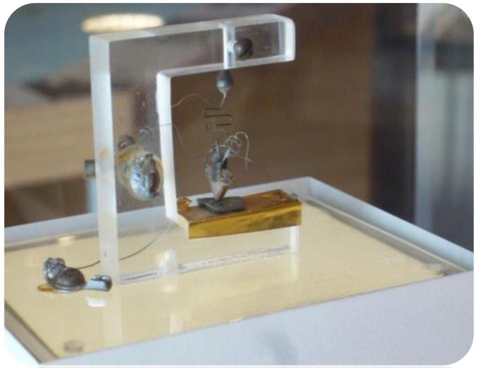
(1925)

(1935)



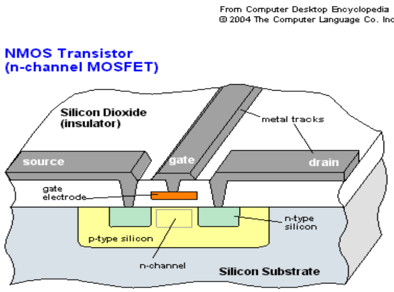
Realization of FET

(1947)



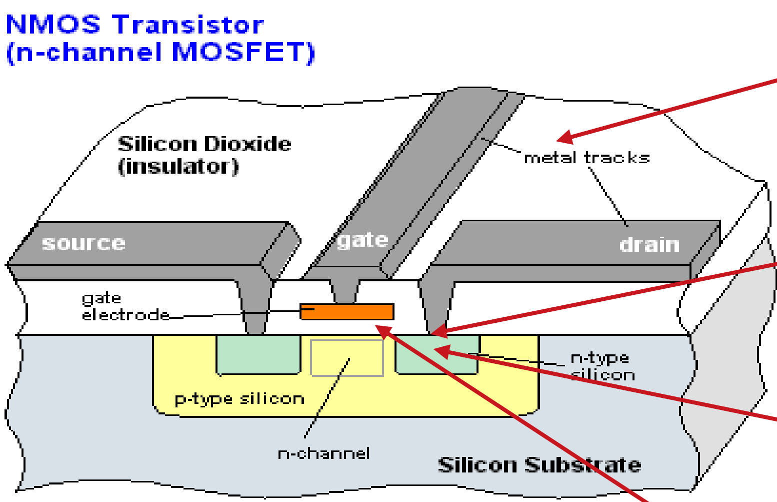
Realization of MOSFET

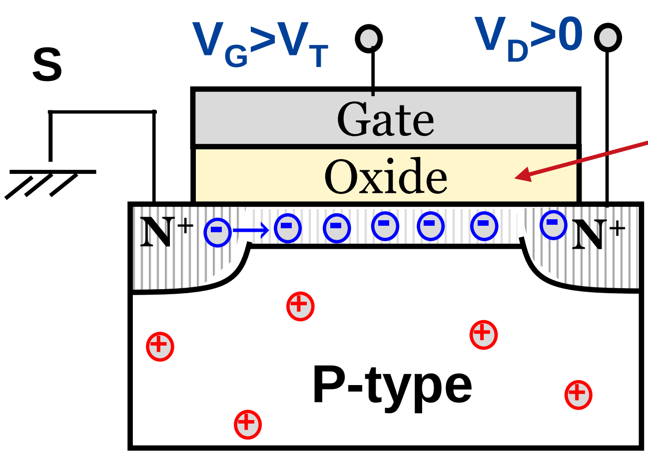
(1959)



3. 1 Why is oxidation of Si needed?

MOSFET





Insulator is needed between electrodes and devices

Si surface should be protected during implantation

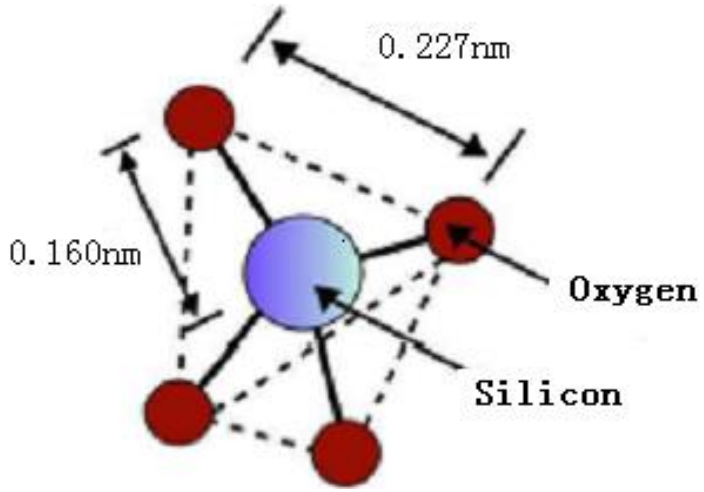
Protected surface should be cleaned for some process

MOS need very very very good oxide and oxide/semiconductor interface

IC should be protected from machinal and chemical damaging.

3. 1 Why is oxidation of Si needed?

- SiO_2 is stable down to 10^{-9} Torr, T>900°C
- SiO_2 is a diffusion barrier for B, P, As
- SiO_2 can be etched with HF which leaves Si unaffected
- SiO_2 is good insulator, Eg>8eV
- SiO_2 has high dielectric breakdown filed, 500V/μm
- SiO_2 grown on Si → clean Si/SiO_2



SiO_2是上帝赐给集成电路的材料（天选之材）

Outline

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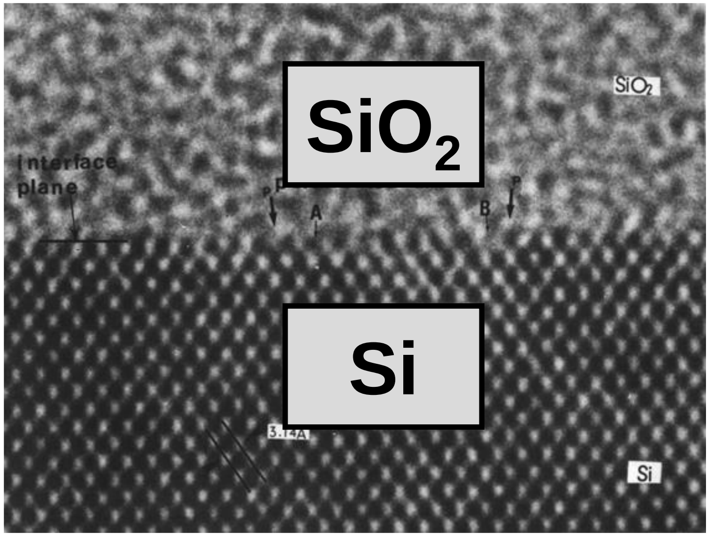
3. 2 Why thermal oxidation is needed?

How can SiO_2 be prepared on Si ?

- **Chemical vapor deposition (CVD)**
- **Physical vapor deposition**
- **Anodic oxidation**
- **Thermal oxidation (best quality)**

3.2.1 Properties of thermal SiO_2



Thermal SiO_2 is amorphous.

Weight Density = 2.20 gm/cm^3

Molecular Density = 2.3E22 molecules/cm^3

Crystalline SiO_2 (Quartz) = 2.65 gm/cm^3

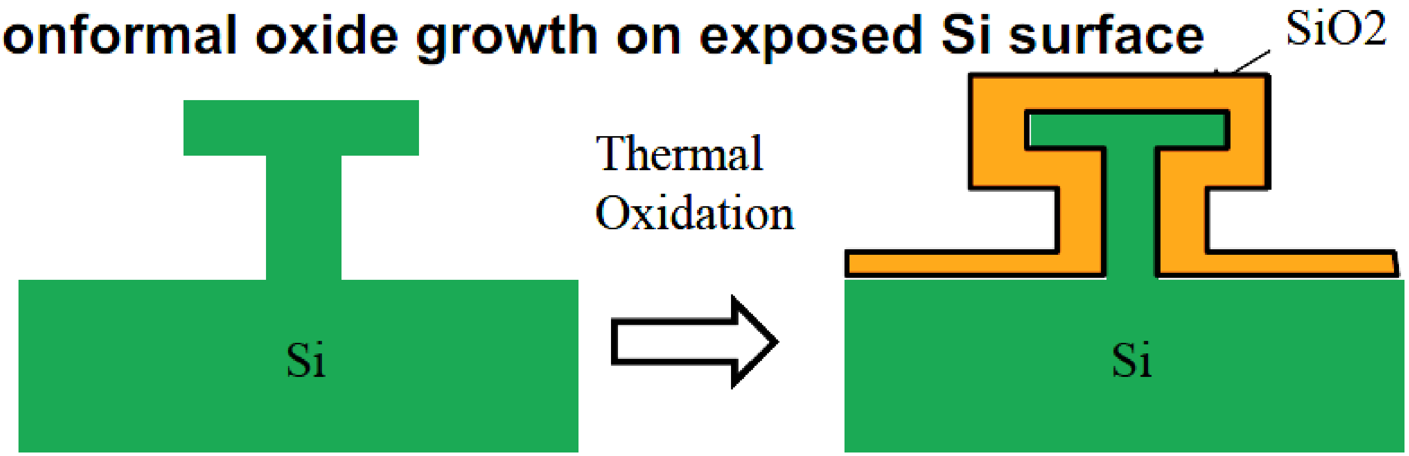
(1) Excellent Electrical Insulator
Resistivity > 10^{20} ohm--cm Energy Gap ~ 9 eV

(2) High Breakdown Electric Field > 10MV/cm

(3) Stable and Reproducible Si/SiO_2 Interface

3.2.1 Properties of thermal SiO_2

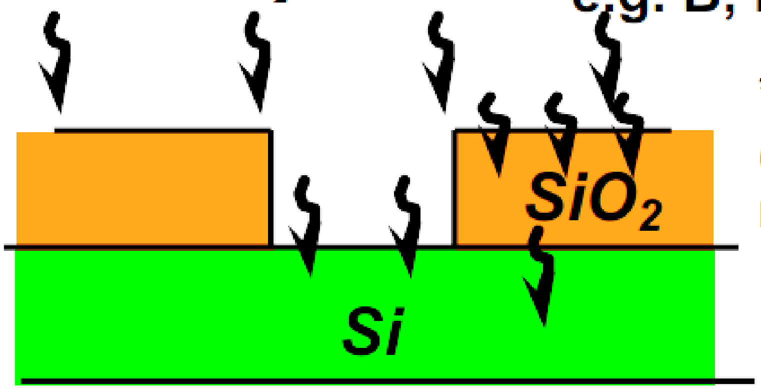
(4) Conformal oxide growth on exposed Si surface



(5) SiO_2 is a good diffusion mask for common dopants

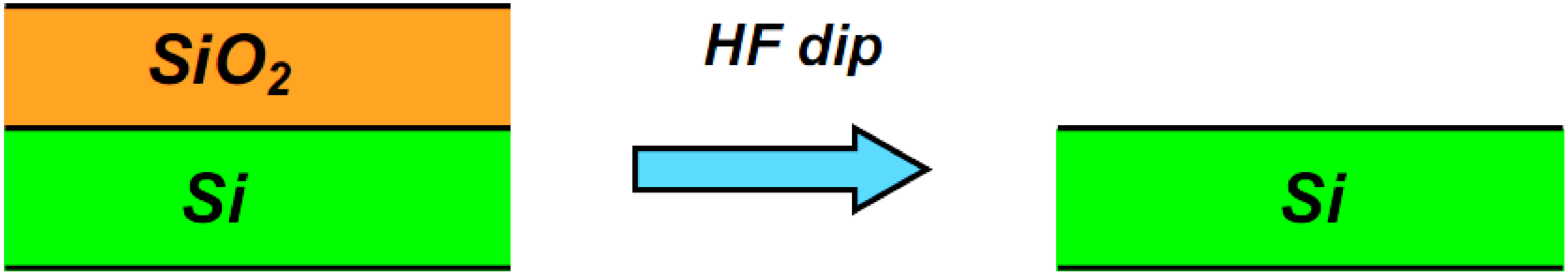
$$D_{sio_2} \ll D_{si}$$

e.g. B, P, As, Sb.

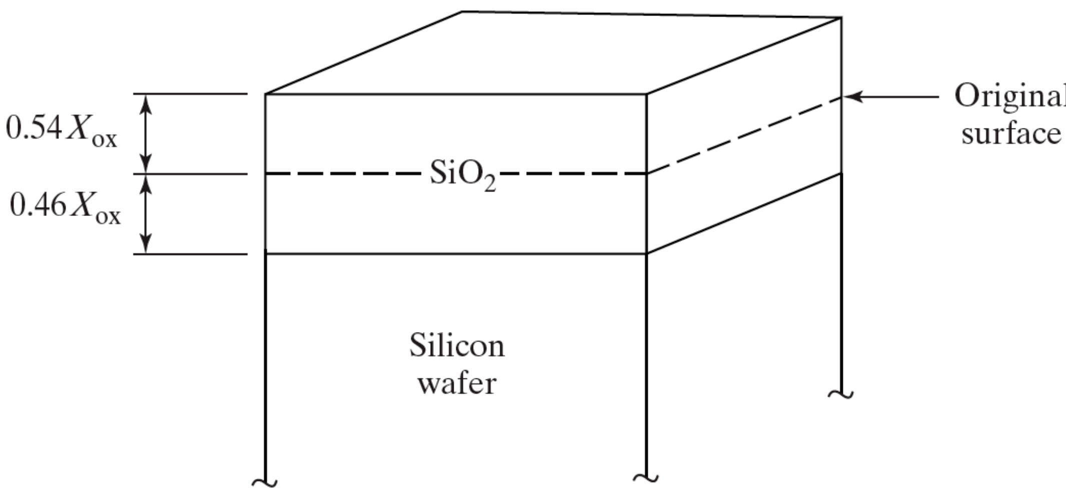


*exceptions are Ga (a p-type dopant) and some metals, e.g. Cu, Au

(6) Very good etching selectivity between Si and SiO_2.



3.2.2 Method of thermal oxidation

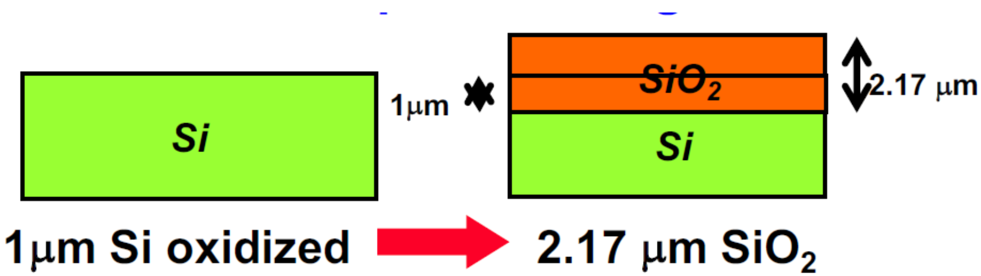


Dry oxidation

$$Si+O_2 \rightarrow SiO_2$$

Wet oxidation

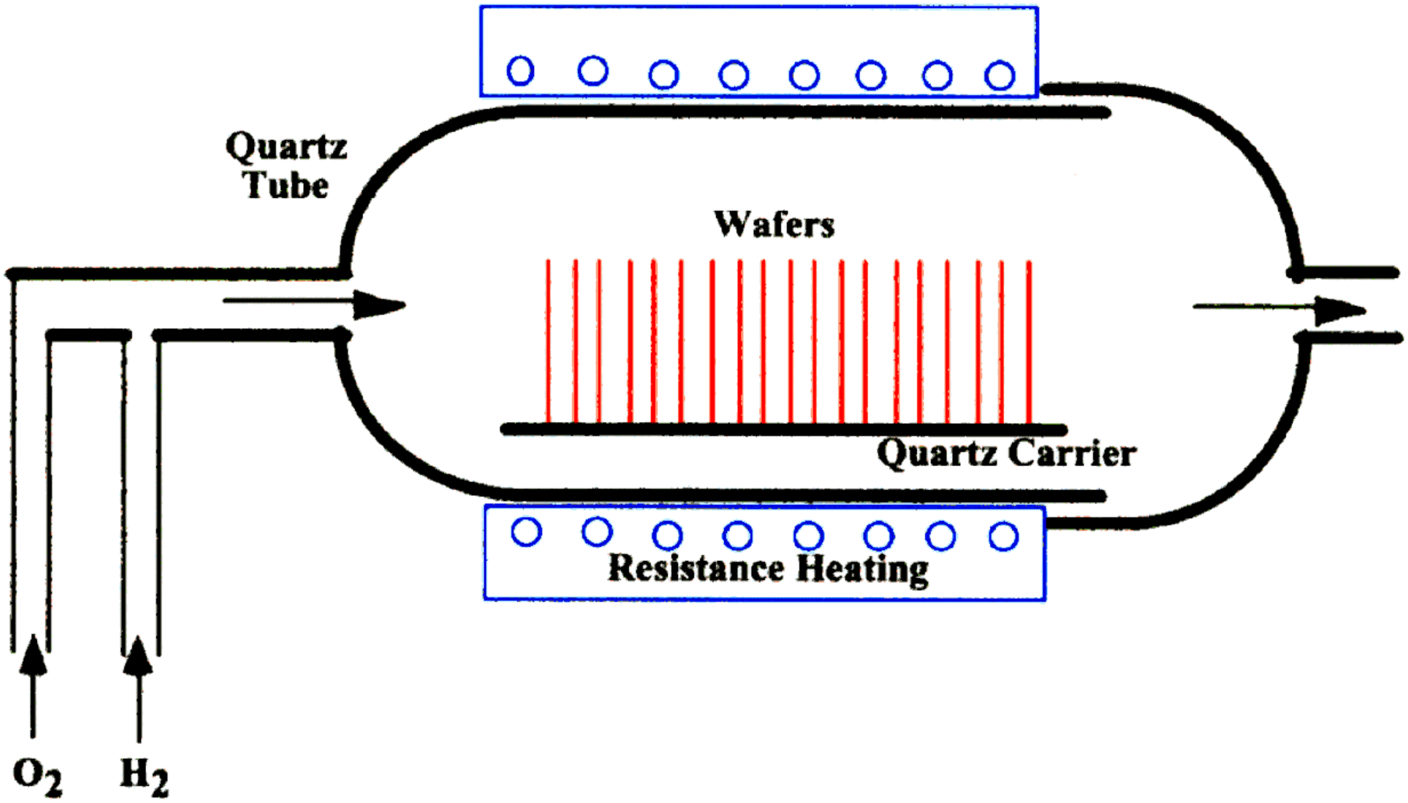
$$Si+2HO_2 \rightarrow SiO_2 + H_2$$



$$\boxed{X_{si}} = X_{ox} \cdot \frac{N_{ox}}{N_{si}}$$

Growth Occurs 54% above and 46% below original surface as silicon is consumed.

3.2.2 Facilities of thermal oxidation



- Oxidation systems are conceptually very simple
- In practice today, vertical furnaces, rapid thermal oxidation (RTO)systems, and fast ramp furnaces are in use

3.2.2 Facilities of thermal oxidation

Horizontal Furnace

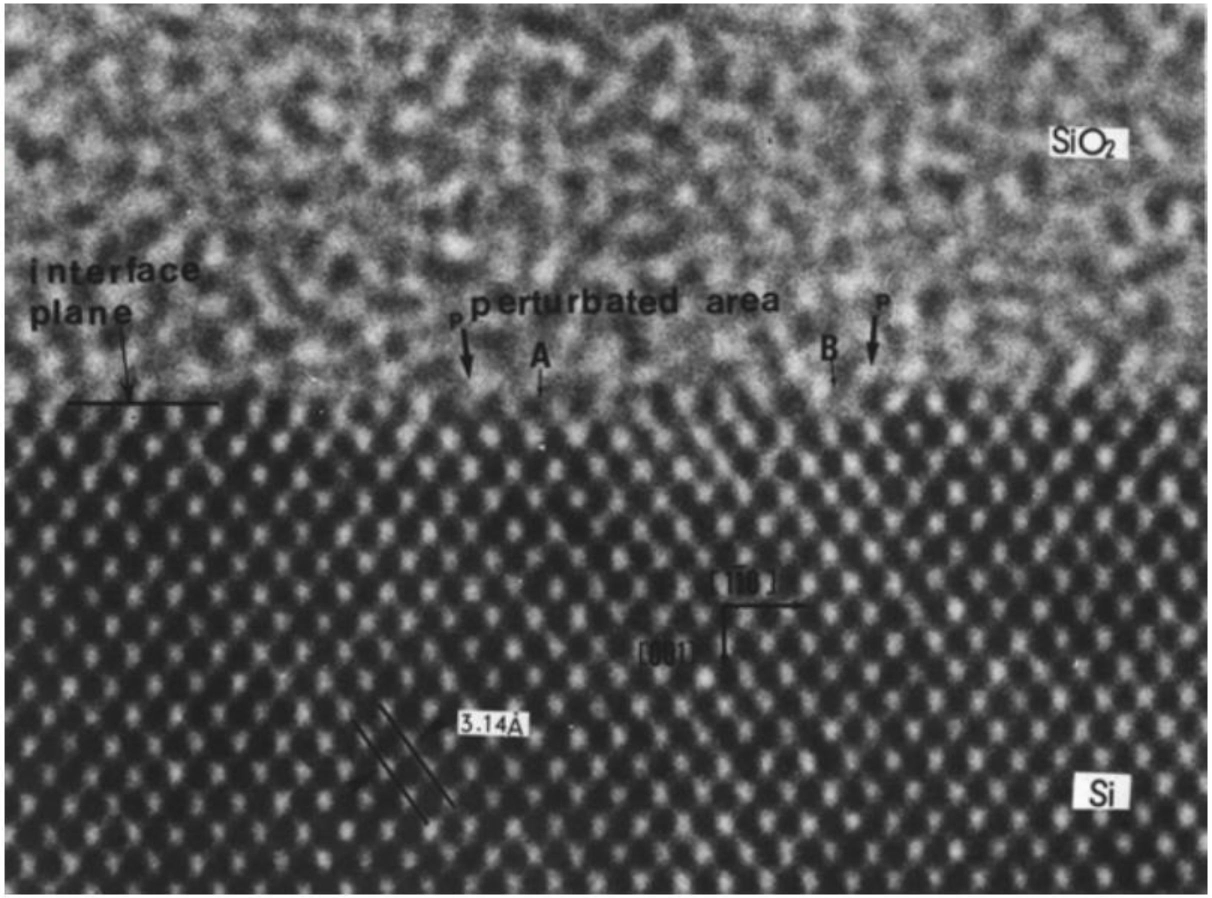
Vertical Furnace

3.2.3 TEM image of Si/SiO_2 interface

Transmission Electron Micrograph

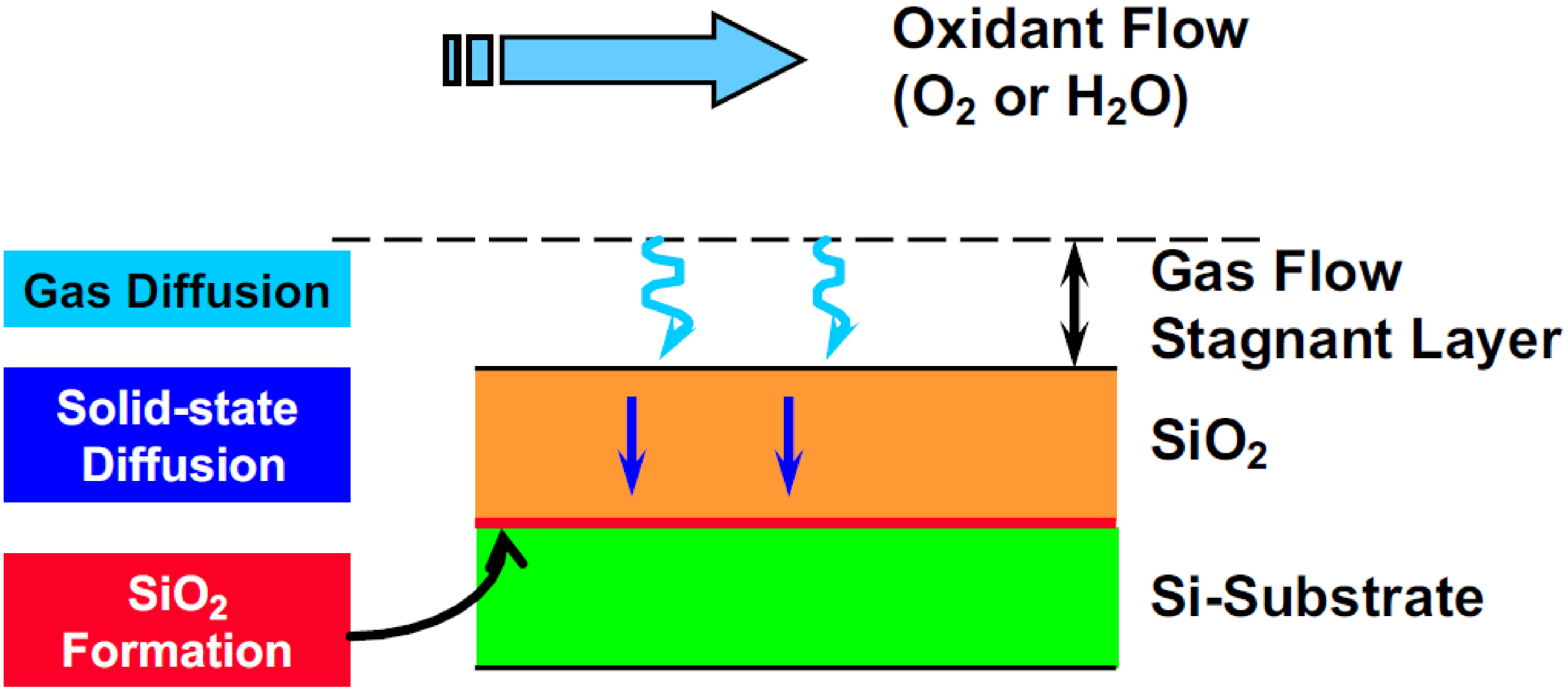
Amorphous SiO_2

Crystalline Si



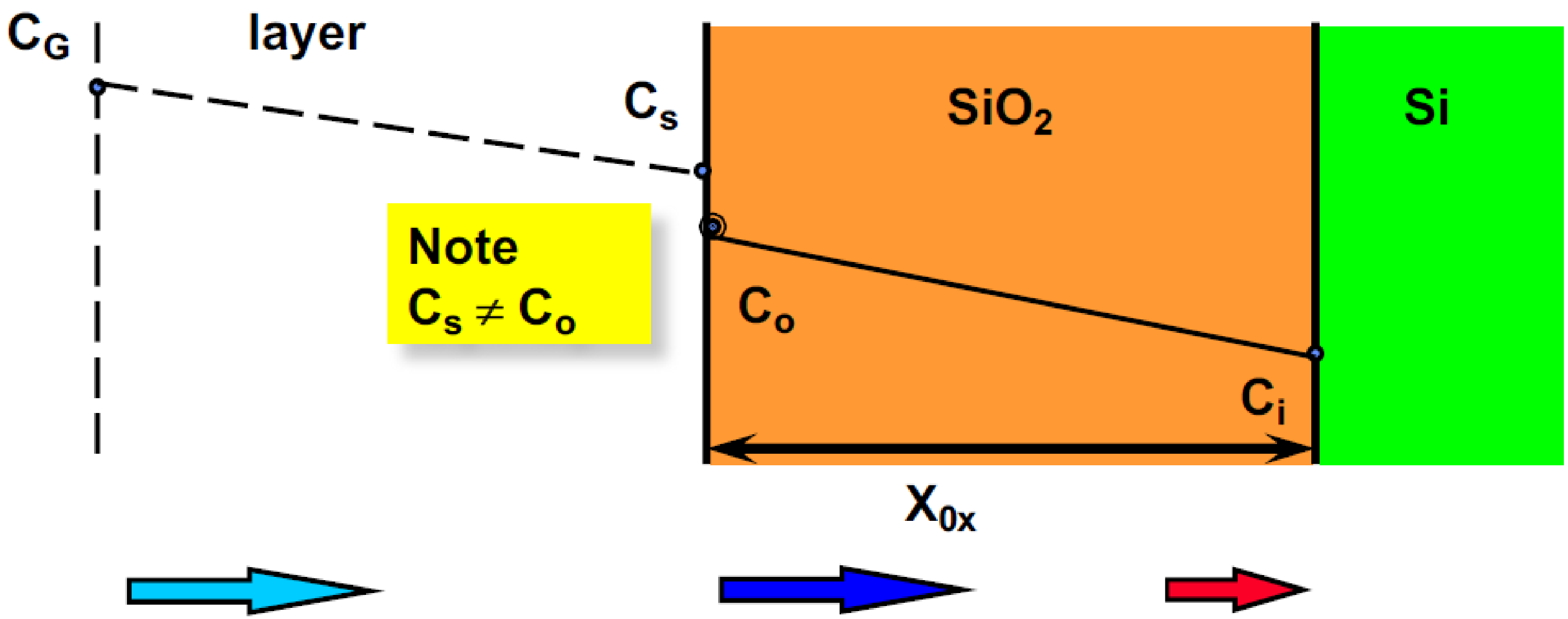
Outline

3.3.1 Kinetics of thermal oxidation



3.3.2 Deal-Grove Model

B. E. Deal and A. S. Grove, J. Appl. Phys. 36, 3771 (1965).

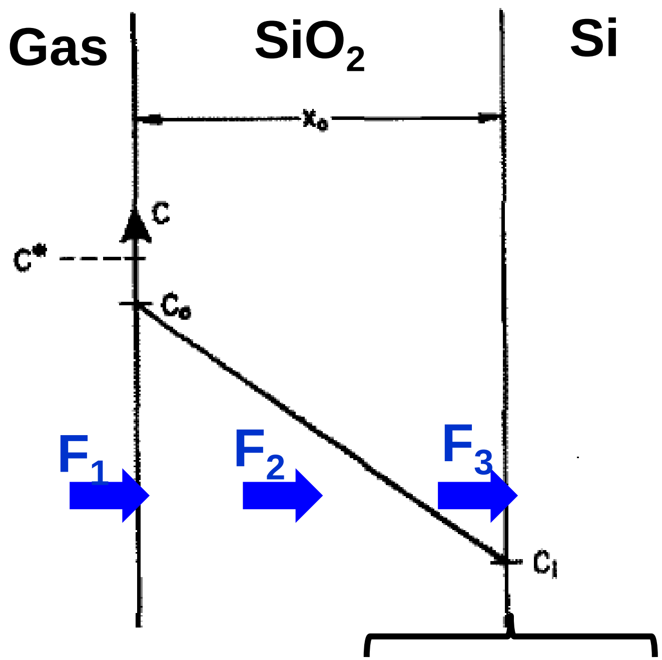


Gas transport | Diffusion through SiO_2 | Reaction at interface

Objective: formulate x=f(t)

3.3.2 Deal-Grove Model-modeling

F: flux – the number of oxygen molecules that crosses a plane per unit area per second



O_2+Si↔SiO_2

Si+2HO_2→SiO_2 +H_2

At surface

$$F_1 = h(C^* - C_o)$$ h_g是质量输运系数

h: Mass transfer coefficient [cm/sec].

In SiO_2 Deff: Diffusivity [cm_2/sec]

$$F_2 = D_{eff}\frac{(C_0 - C_i)}{x}$$ 菲克第一定律

At SiO_2/Si interface

$$F_3 = kC_i$$

k: Oxidation reaction rate constant

Oxidant diffusion→ interface reaction.

In SiO_2 is growth , the Si at the interface is changed to SiO_2. Thus, the flux at the interface is :

Oxidation rate

$$-\frac{dx}{dt}=\frac{F}{N_1}$$

N1: oxidant molecules/unit volume required to form a unit volume of SiO_2. The molecular density of SiO_2 is $2.2\times10^{22}/cm^3$. In oxidation，to get a SiO_2 molecular，we need a O_2 molecular in dry oxidation or two H_2O moleculars in wet oxidation.

3.3.2 Deal-Grove Model-modeling

C^* is a controlled process variable proportional to the O_2 pressure

Only Co and Ci are the 2 unknown variables

At steady state

$$F_1 = F_2 = F_3$$

$$C_i = \frac{C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

$$C_o = \frac{(1 + \frac{kx}{D_{eff}})C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

- **When $D_{eff} << kx$,**

  $C_i \sim 0$, $C_o \sim C^*$,

oxidation is mainly controlled by diffusion.

- **When D_{eff} is very large，$C_i = C_o = C^*/(1+k/h)$,**

oxidation is mainly controlled by reaction.

3.3.2 Deal-Grove Model-oxidation rate

In SiO_2 is growth, the Si at the interface is changed to SiO_2. Thus, the flux at the interface is:

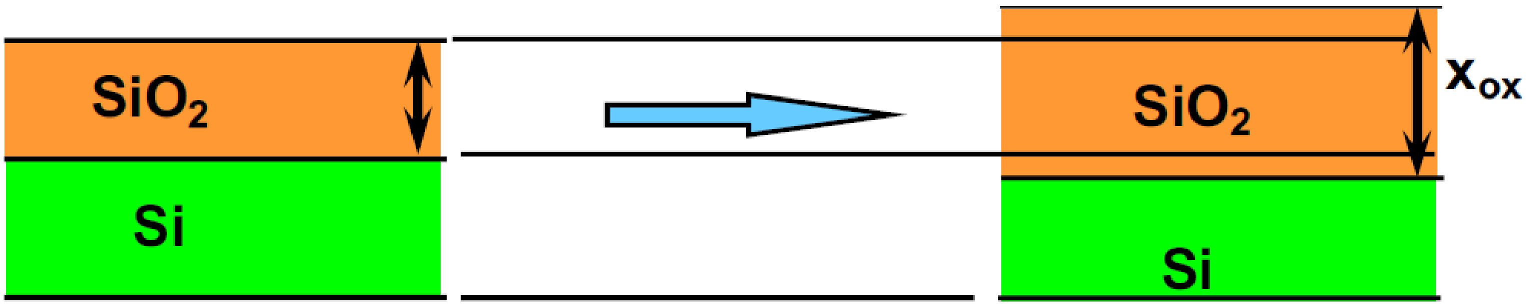
Oxidation rate

$$-\frac{dx}{dt} = \frac{F}{N_1}$$

$$F = N_1 \frac{\mathrm{dx}}{\mathrm{dt}} = kC_i = k\frac{C^*}{1+\frac{k}{h}+\frac{kx}{D_{eff}}}$$

N1: oxidant molecules/unit volume required to form a unit volume of SiO_2. The molecular density of SiO_2 is $2.2\times10^{22}/cm^3$. In oxidation, to get a SiO_2 molecular, we need a O_2 molecular in dry oxidation or two H_2O moleculars in wet oxidation.

Initial Condition: At t = 0 , X = X₀



$$x_0^{\ 2} - x^2 + A(x_0 - x) = Bt$$

$$x^2 + Ax = B(t + \tau)$$

$$A \equiv 2D_{eff}\left(\frac{1}{k} + \frac{1}{h}\right) \qquad B \equiv 2D_{eff}\frac{C^*}{N_1}$$

$$\tau = x_0^{\ 2} + Ax_0$$

3.3.2 Deal-Grove Model-modeling

C^* is a controlled process variable proportional to the O_2 pressure

Only Co and Ci are the 2 unknown variables

At steady state

$$F_1 = F_2 = F_3$$

$$C_i = \frac{C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

$$C_o = \frac{\left(1 + \frac{kx}{D_{eff}}\right)C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

- **When $D_{eff} << kx$,**
  $C_i \sim 0$, $C_o \sim C^*$,

oxidation is mainly controlled by diffusion.

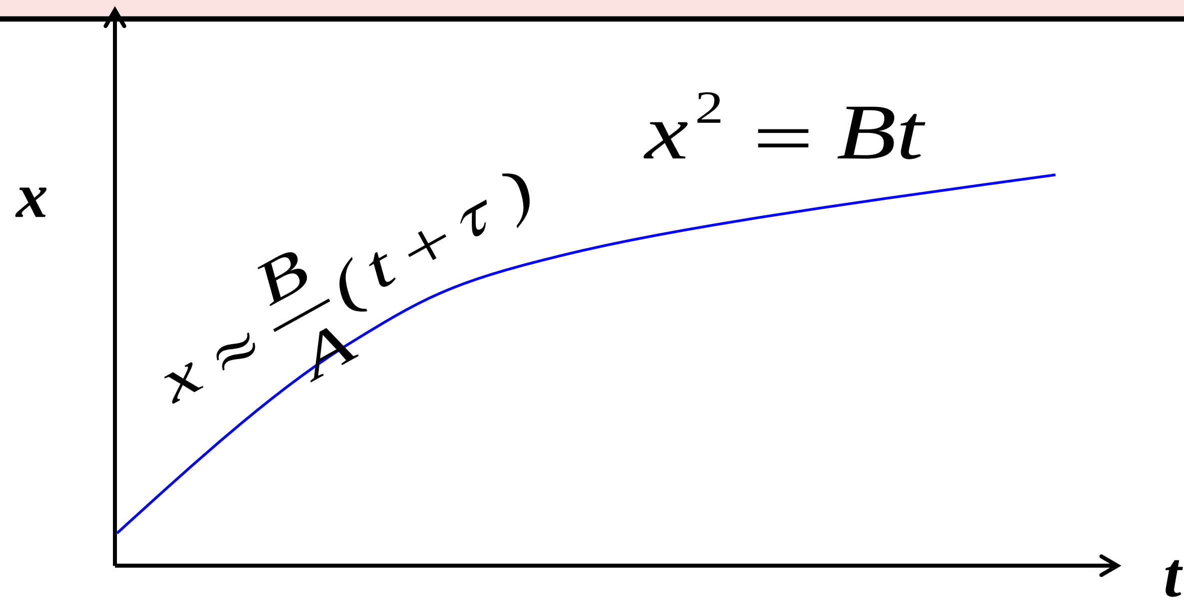
- **When D_{eff} is very large, $C_i = C_o = C^*/(1+k/h)$,**

oxidation is mainly controlled by reaction.

3.3.2 Deal-Grove Model-two stage of oxidation

$$x_i = \frac{A}{2}\left(\sqrt{1+\frac{t+\tau}{A^2/4B}}-1\right)$$

When $t << A^2/4B$时，	When $t >> A^2/4B$，
$x \approx \frac{B}{A}(t+\tau)$	$x^2 \approx B(t+\tau)$
（线性规律，反应控制）	（抛物型规律，扩散控制）



$\sqrt{1+x}-1 \sim \frac{x}{2}$

3.3.2 Deal-Grove Model-rate constants

Rate constants B and B/A have physical meaning: oxidant diffusion and interface reaction rate respectively

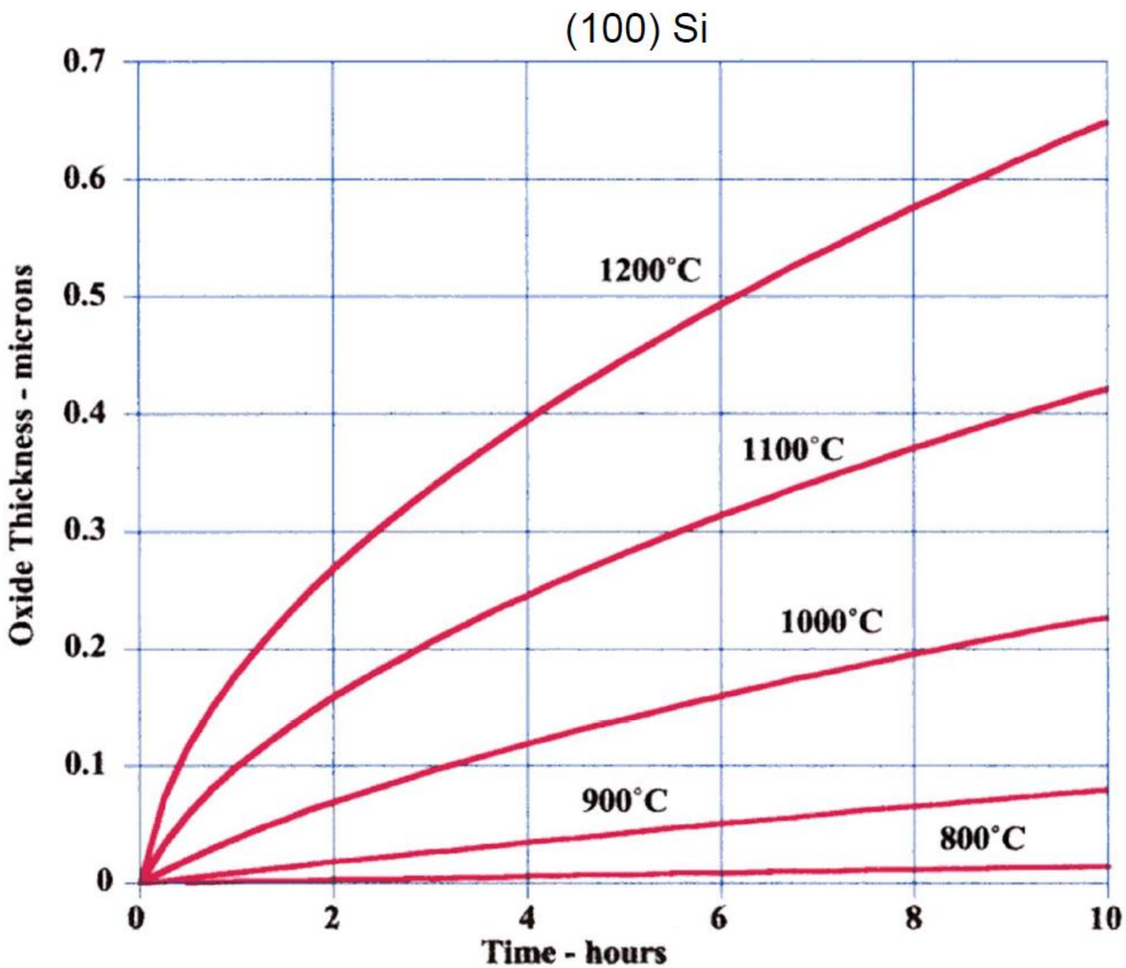
$$B = C_1 \exp(-E_1/kT) \qquad B \equiv 2D_{Sio2}\frac{C^*}{N_1}$$

$$\frac{B}{A} = C_2 \exp(-E_2/kT) \qquad A \equiv 2D_{Sio2}\left(\frac{1}{k}+\frac{1}{h}\right)$$

Ambient	B	B/A
Dry O_2	$C_1 = 7.72 \times 10^2\ \mu^2\ hr^{-1}$ $E_1 = 1.23$ eV	$C_2 = 6.23 \times 10^6\ \mu\ hr^{-1}$ $E_2 = 2.0$ eV
Wet O_2	$C_1 = 2.14 \times 10^2\ \mu^2\ hr^{-1}$ $E_1 = 0.71$ eV	$C_2 = 8.95 \times 10^7\ \mu\ hr^{-1}$ $E_2 = 2.05$ eV

- activation energy for B varies with process (diffusivity of different species through the oxide)
- activation energy for B/A is roughly constant (surface reaction rate)

3.3.2 Deal-Grove Model-simulation



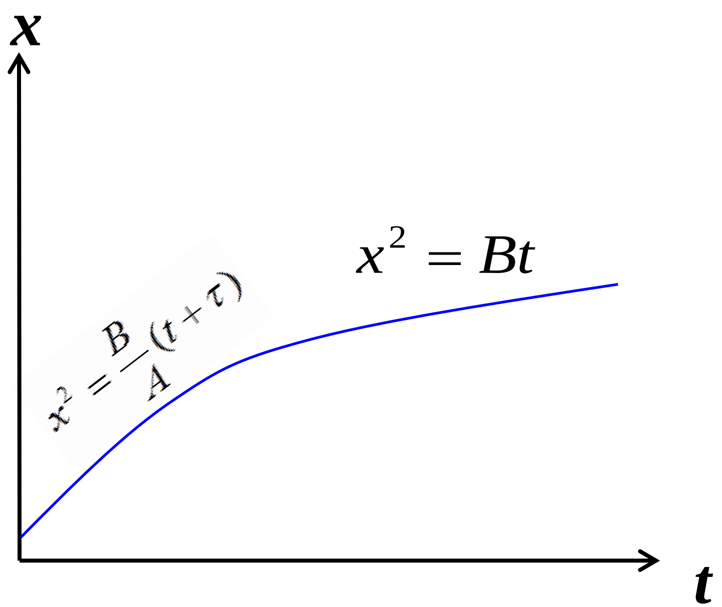
- **Calculated dry O_2 oxidation rates using Deal Grove.**
  - linear regime followed by parabolic behavior
  - dry oxidations typically used for oxide thicknesses < 100 to 200 nm

3.3.3 Factors influencing oxidation rate

- **Partial pressure of oxidant (氧化剂分压)**
- **Temperature of oxidation (氧化温度)**
- **Surface orientation of Si （硅晶面取向）**
- **Doping and Impurity （杂质）**
- **Dry and Wet oxidation (干湿法氧化）**

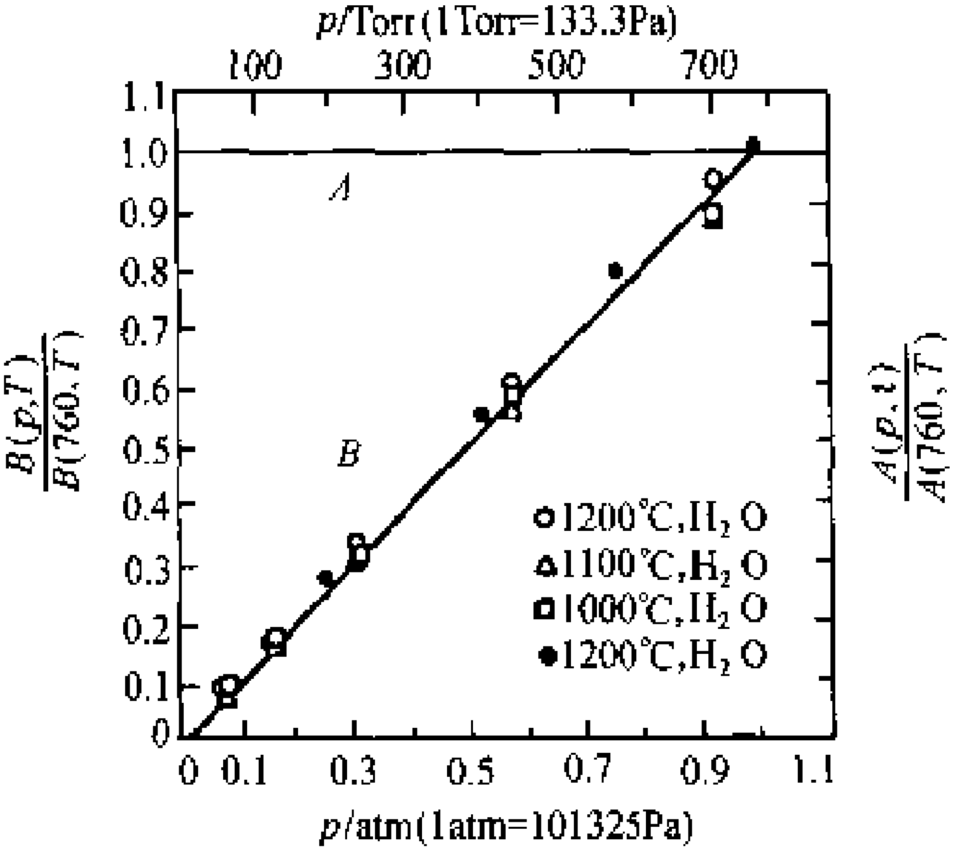
3.3.3 Factors influencing oxidation rate (1)

Effect of Partial pressure of oxidant (氧化剂分压影响)



$$B \equiv 2D_{SiO2}\frac{C^*}{N_1}$$

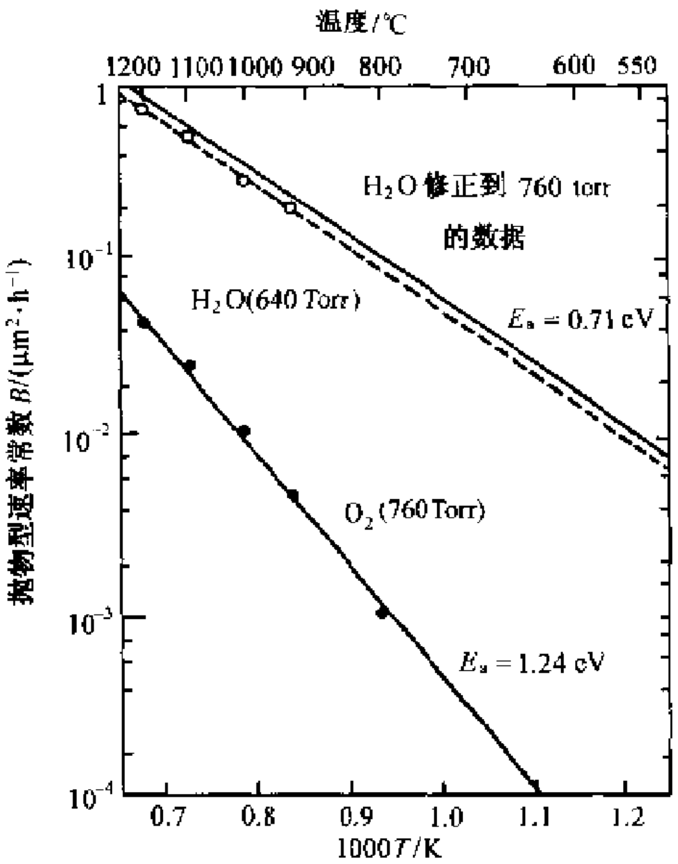
$$C^* = Hp_g$$

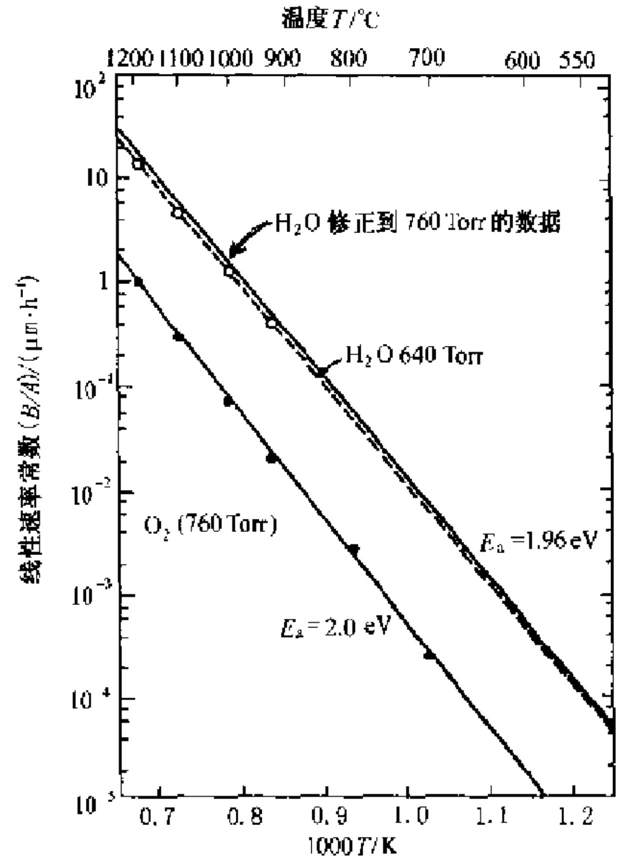


P ↑ , C* ↑ ➡ B ↑

Effect of temperature of oxidant (温度影响)

$$B \equiv 2\boxed{D_{SiO2}}\frac{C^*}{N_1} \qquad A \equiv 2D_{SiO2}\left(\frac{1}{\boxed{k}}+\frac{1}{h}\right)$$

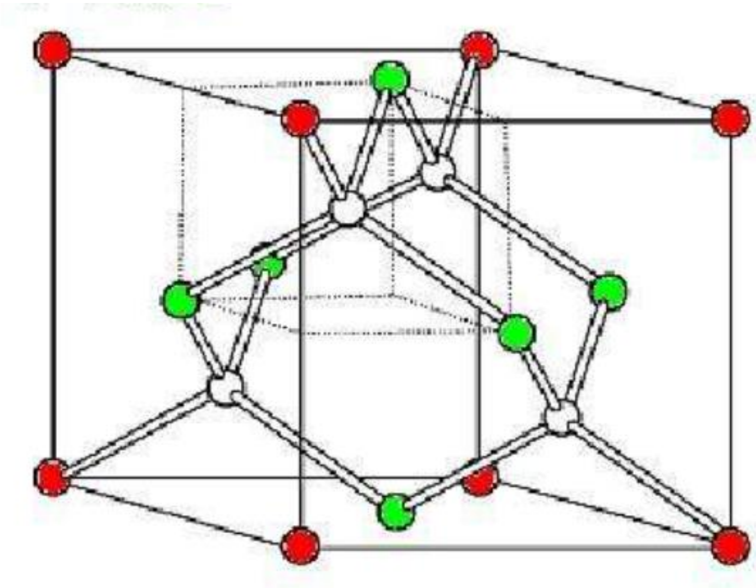


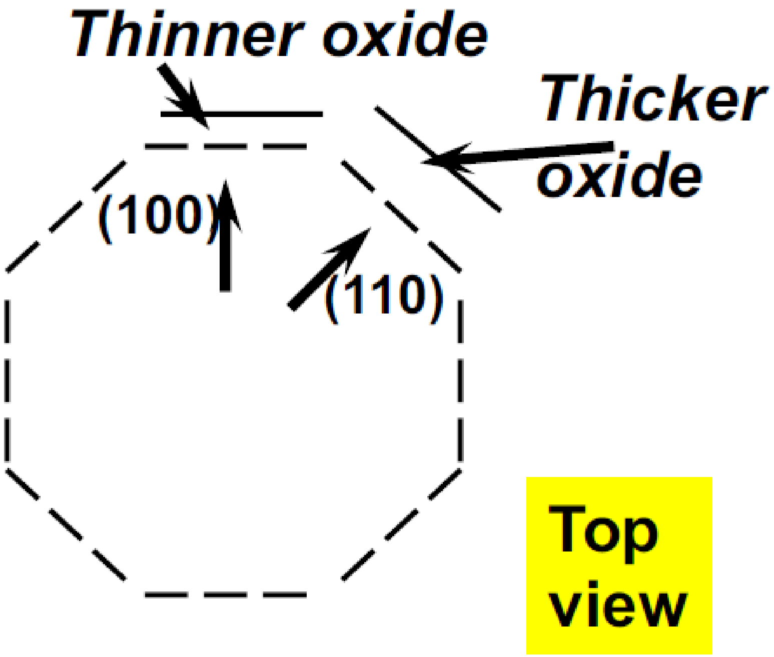


T↑，D_{SiO2}↑，K_s↑ ➡ B↑，B/A↑

3.3.3 Factors influencing oxidation rate (3)

Effect of surface orientation (硅晶体取向影响)



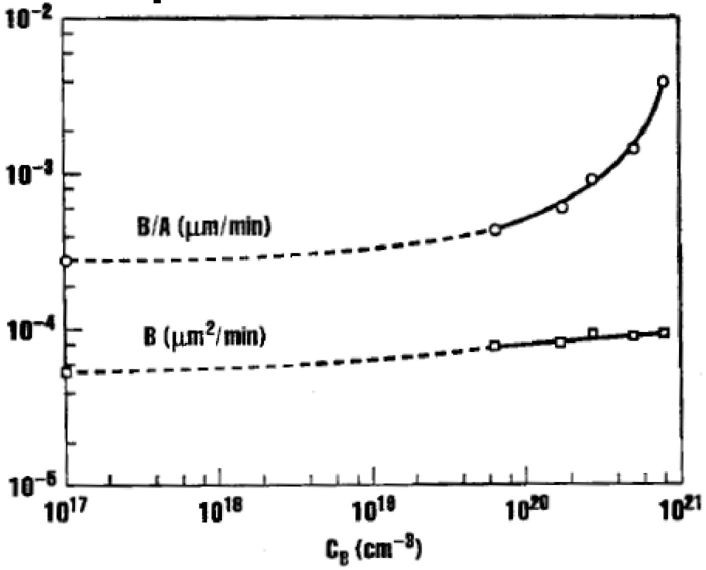


$$A \equiv 2D_{SiO2}\left(\frac{1}{\boxed{k}}+\frac{1}{h}\right)$$

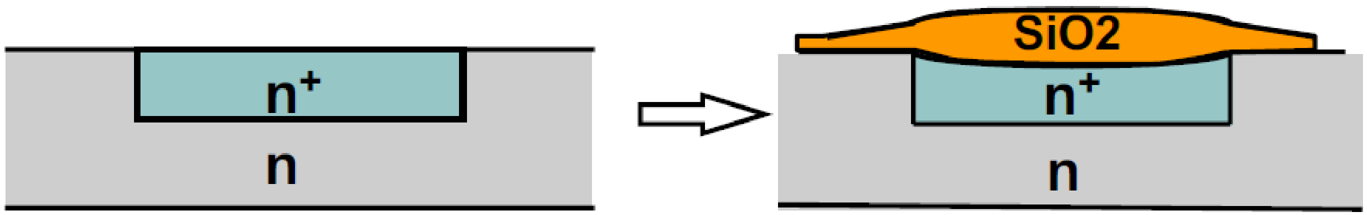
(111) >(110)>(100)

Effect of doping (掺杂浓度影响)

Coefficients for dry oxidation at 900ºC as function of surface Phosphorus concentration



Dry oxidation, 900°C



3.3.3 Factors influencing oxidation rate (5)

Dry / Wet Oxidation

Note : "dry" and "wet" oxidation have different N1 factors

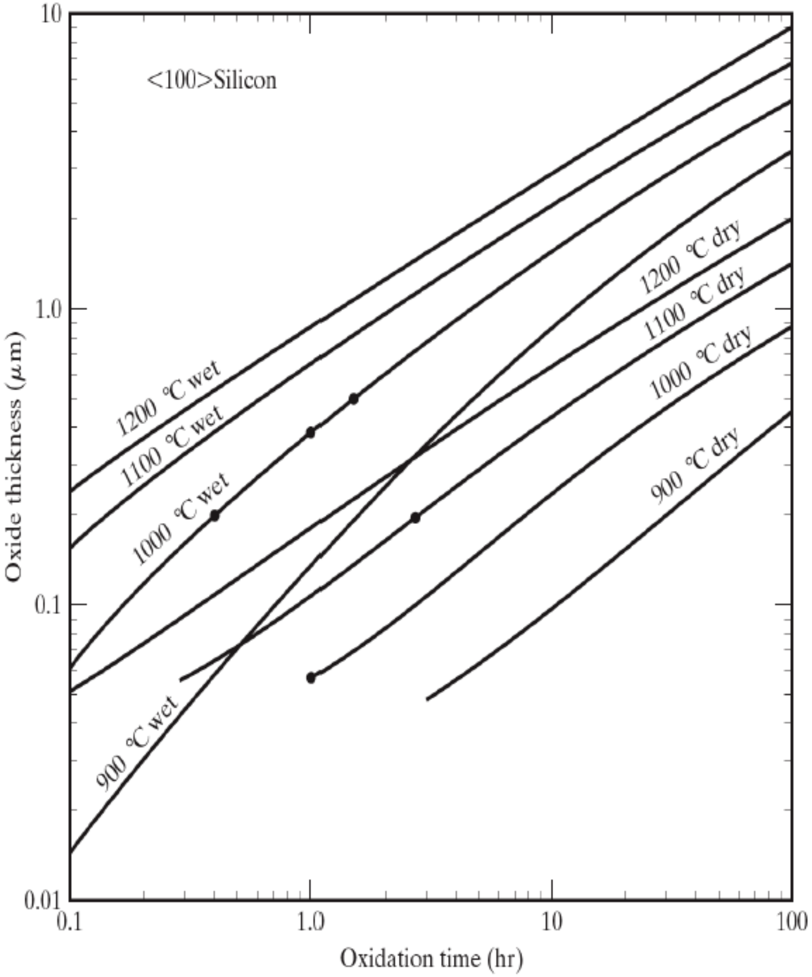
$$N_1 = \boxed{2.3 \times 10^{22} \ / \ cm^3}$$ ***for O_2 as oxidant***

$$Si + O_2 \rightarrow SiO_2$$

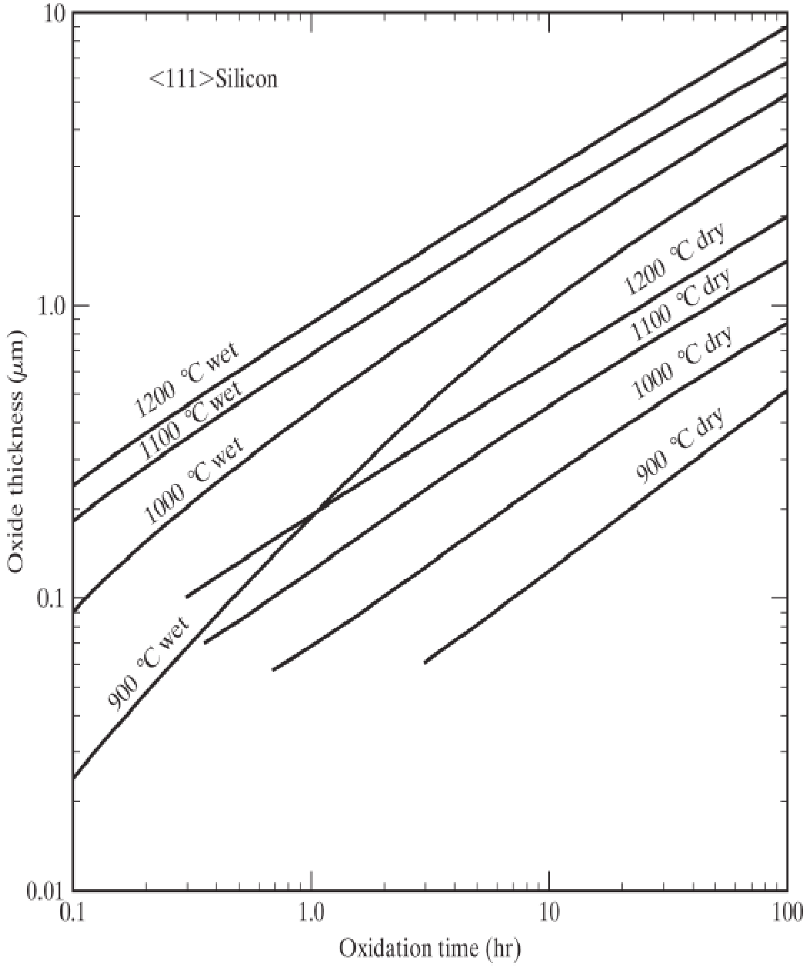
$$N_1 = \boxed{4.6 \times 10^{22} \ / \ cm^3}$$ ***for H_2O as oxidant***

$$Si + 2H_2O \rightarrow SiO_2 + 2H_2 \uparrow$$

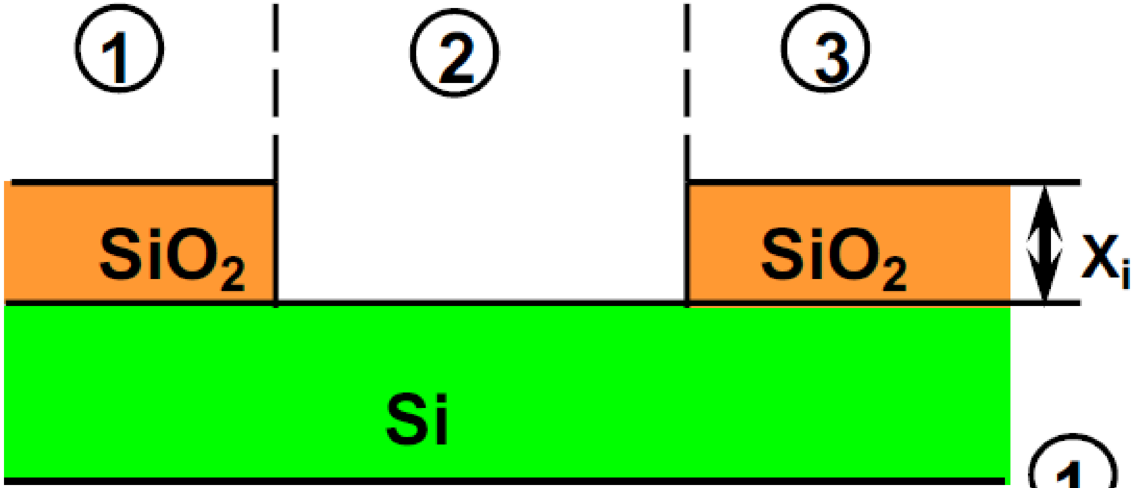
Thermal Oxidation on <100> Silicon



Thermal Oxidation on <111> Silicon

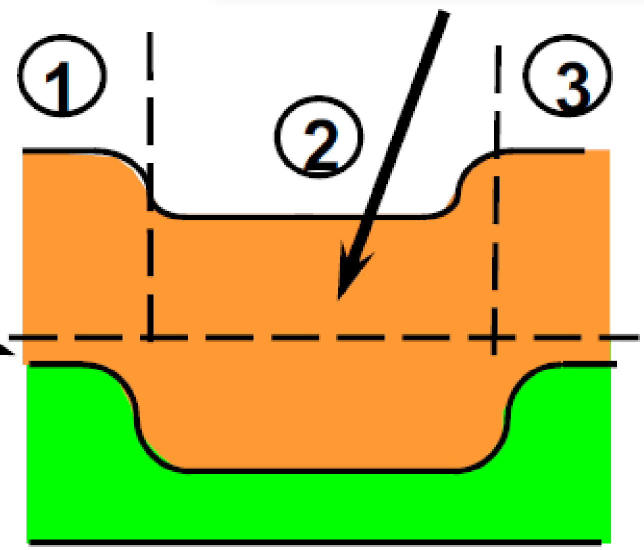


Effect of X_0 on wafer Topography



more oxide grown
more Si consumed

less oxide grown
less Si consumed



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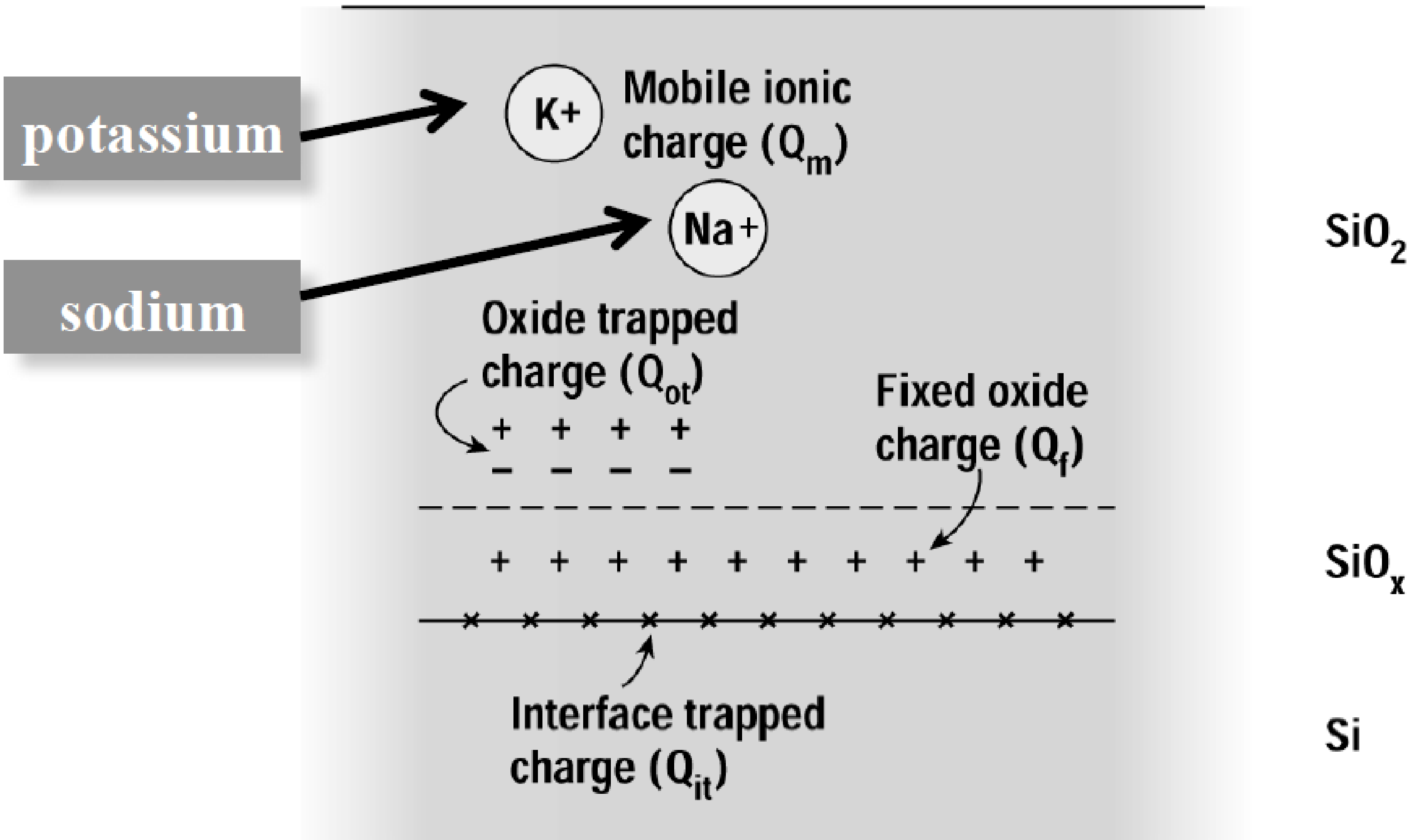
3.4.1 Thermal oxide charges

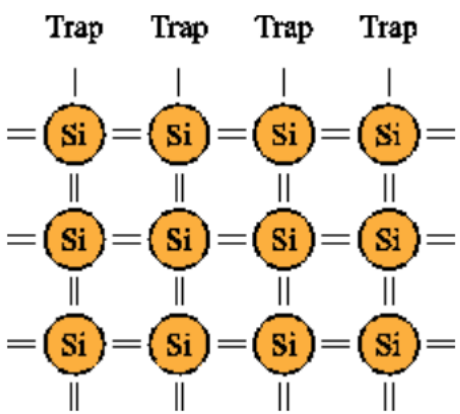


3.4.1 Oxide Quality improvement

To minimize Interface Charges Q_f and Q_{it}

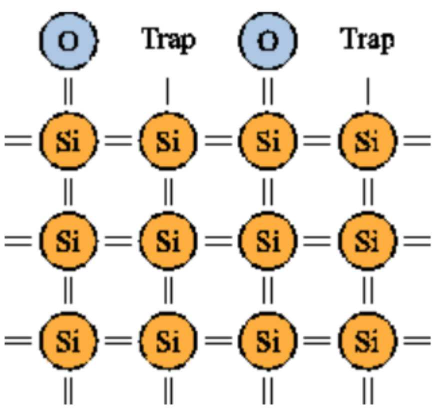
- **Use inert gas ambient (Ar or N_2) when cooling down at end of oxidation step**

- **A final annealing step at 400--450$_o$C is performed with 10%H_2+90%N_2 ambient ("*forming gas*") after the IC metallization step.**

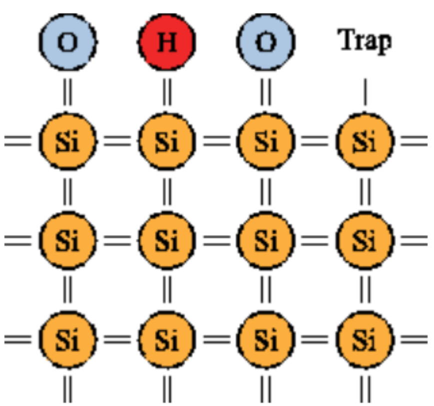


Silicon surface

10^{15} atom/cm^{-2}



Unpassivated Si/SiO_2



After hydrogen passivation

D_{it}: 10^{10}-10^{12} atom/cm^{-2}

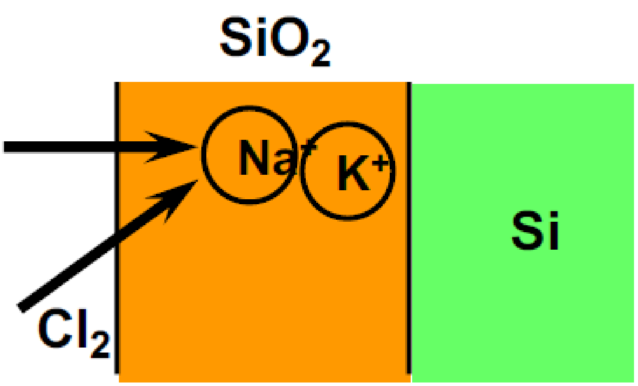
3.4.2 Oxide Quality improvement

Introduction of halogen species during oxidation

e.g. **add *~1-- 5%* HCl or TCE (trichloroethylene) to O_2**

-**reduction in metallic contamination**

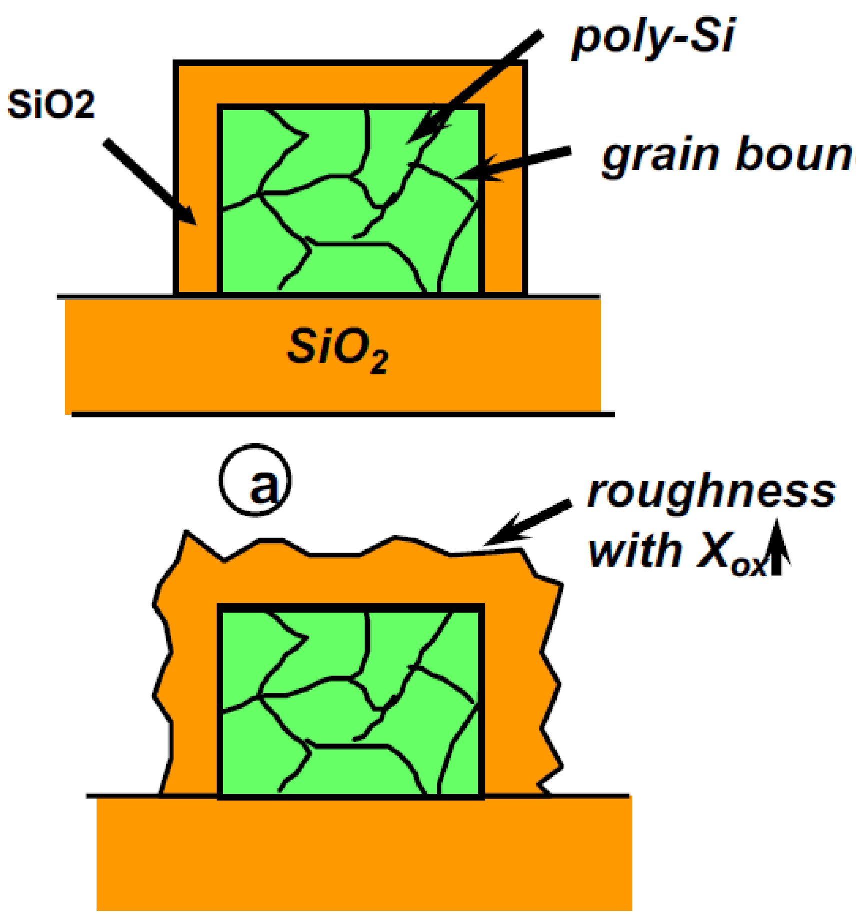
-**improved SiO_2/Si interface properties**

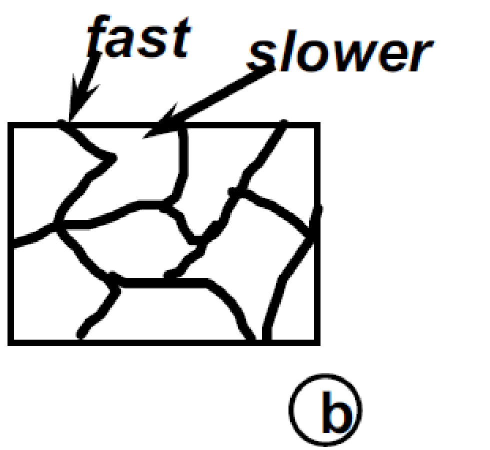


$$M + Cl \rightarrow MCl$$

Na^+ or K^+ in SiO_2 are mobile!

3.4.3 Polycrystalline Si Oxidation





Overall growth rate is higher than single-crystal Si

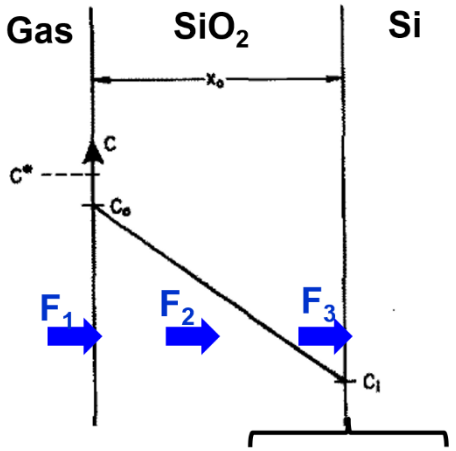
Outline

3.5.1 Summary

- **Why is oxidation needed?** **Functiond and advantage for ICs**
- **Method of thermal oxidation**

  Dry oxidation **Wet oxidation**

  $Si+O_2 \rightarrow SiO_2$ $Si+2HO_2 \rightarrow SiO_2 + H_2$



$O_2+Si \leftrightarrow SiO_2$

$Si+2HO_2 \rightarrow SiO_2 + H_2$

- **Thermal oxidation kinetics**

Deal-Grove model

- **Influencing factors of oxidation rate**

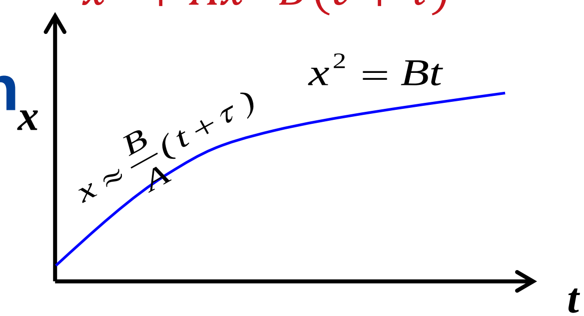
Oxidant pressure, temperature

Surface orientation, doping concentration

Dry and wet method, initial thickness

- **Other aspects**

$$x^2 + Ax = B(t+\tau)$$



3.5.2 Homework

MANY suggestions have been made in the literature to explain the initial rapid oxidation kinetics. None have been widely accepted.
However, accurate mathematical expressions have been generated and are used in simulators (e.g. SUPREM IV and Sprocess), based upon some of these models.
Here, we will consider three of the physical models:

1. Reisman et al., Model
2. Han and Helms Model
3. Massoud et al., Model

搜索文献了解其中一种model，并用中文简述其核心内容

谢 谢